EXP NO: 5	*CLUSTERING WITH K-MEANS AND DIMENSIONALITY REDUCTION WITH PCA*

AIM:

To implement the K-Means clustering algorithm on the Wine dataset without using class labels, find the optimal number of clusters, and apply Principal Component Analysis (PCA) for dimensionality reduction to study its effect on the clusters.

ALGORITHM:

1. Import the required libraries (numpy, pandas, matplotlib, scikit-learn).
2. Load the Wine dataset (13 features) and scale the features using StandardScaler (class

   labels are not used for training).
3. Run K-Means for k = 2 to 10 and record the inertia and silhouette score; plot the Elbow and Silhouette curves to choose k.
4. Build the K-Means model with k = 3 on the original 13 features and evaluate it using silhouette score and adjusted Rand index.
5. Apply PCA and plot the scree plot and cumulative explained variance; find the number of components that retain 95% of the variance.
6. Run K-Means again on the PCA-reduced data (95% variance and 2 components) and compare the results with the original data.
7. Visualise the clusters and centroids in the 2-D PCA space.

CODE:

```
# 1. Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score, adjusted_rand_score


# 2. Load dataset (13 chemical features; class labels are NOT used for training)
wine = load_wine()
X, y_true = wine.data, wine.target
X_scaled = StandardScaler().fit_transform(X)
print("Data shape:", X_scaled.shape)
```

```
# 3. Find the optimal number of clusters (Elbow + Silhouette)
inertia, sil = [], []
K = range(2, 11)
for k in K:
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X_scaled)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(X_scaled, km.labels_))
    print(f"k={k}  Inertia={km.inertia_:.2f} Silhouette={sil[-1]:.4f}")

plt.figure(figsize=(11, 4))
plt.subplot(1, 2, 1)
plt.plot(K, inertia, 'bo-'); plt.xlabel("Number of clusters (k)");
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.subplot(1, 2, 2)
plt.plot(K, sil, 'gs-'); plt.xlabel("Number of clusters (k)");
    plt.ylabel("Silhouette Score")
plt.title("Silhouette Analysis")
plt.tight_layout()
plt.show()

# 4. K-Means with k = 3 on the full 13-dimensional data
kmeans = KMeans(n_clusters=3, n_init=10, random_state=42)
labels_full = kmeans.fit_predict(X_scaled)
print("\n--- K-Means on original 13 features ---")
print("Cluster sizes:", np.bincount(labels_full))
print("Silhouette Score:", round(silhouette_score(X_scaled, labels_full), 4))
print("Adjusted Rand Index:", round(adjusted_rand_score(y_true, labels_full),
    4))

# 5. Dimensionality reduction with PCA
pca_full = PCA().fit(X_scaled)
explained = pca_full.explained_variance_ratio_
cumulative = np.cumsum(explained)
print("\nExplained variance ratio (first 5 components):",
    np.round(explained[:5], 4))
print("Cumulative variance (first 5 components)     :",
    np.round(cumulative[:5], 4))
n_95 = int(np.argmax(cumulative >= 0.95) + 1)
print("Components needed to retain 95% variance:", n_95)

plt.figure(figsize=(11, 4))
plt.subplot(1, 2, 1)
plt.bar(range(1, 14), explained, color='steelblue')
plt.xlabel("Principal Component"); plt.ylabel("Explained Variance Ratio");
    plt.title("Scree Plot")
plt.subplot(1, 2, 2)
plt.plot(range(1, 14), cumulative, 'ro-')
plt.axhline(0.95, color='gray', linestyle='--', label='95% variance')
plt.xlabel("Number of Components"); plt.ylabel("Cumulative Variance")
plt.title("Cumulative      Explained      Variance");      plt.legend()
plt.tight_layout()
plt.show()

# 6. K-Means after PCA (95% variance, and 2 components)
```

```
X_pca95 = PCA(n_components=n_95).fit_transform(X_scaled)
X_pca2 = PCA(n_components=2).fit_transform(X_scaled)
labels_95 = KMeans(n_clusters=3, n_init=10,
    random_state=42).fit_predict(X_pca95)
km2 = KMeans(n_clusters=3, n_init=10, random_state=42).fit(X_pca2)
labels_2 = km2.labels_

results = pd.DataFrame({
    "Data used": ["Original (13 features)", f"PCA ({n_95} components, 95%)",
        "PCA (2 components)"],
    "Silhouette": [silhouette_score(X_scaled, labels_full),
        silhouette_score(X_pca95, labels_95),
                   silhouette_score(X_pca2, labels_2)],
    "Adjusted Rand Index": [adjusted_rand_score(y_true, labels_full),
        adjusted_rand_score(y_true, labels_95),
                            adjusted_rand_score(y_true, labels_2)]})
print("\nK-Means before and after PCA:\n")
print(results.round(4).to_string(index=False))
print("\nCluster vs Wine-class cross-tab (PCA 2 components):\n",
      pd.crosstab(labels_2, y_true, rownames=['Cluster'], colnames=['Class']))

# 7. Visualise clusters in the 2-D PCA space
plt.figure(figsize=(7, 5))
plt.scatter(X_pca2[:, 0], X_pca2[:, 1], c=labels_2, cmap='viridis', s=40,
    edgecolor='k')
plt.scatter(km2.cluster_centers_[:, 0], km2.cluster_centers_[:, 1], c='red',
    marker='X', s=250, label='Centroids')
plt.xlabel("PCA 1"); plt.ylabel("PCA 2")
plt.title("K-Means Clusters (k=3) in PCA Space")
plt.legend()
plt.show()
```

OUTPUT:

```
Data shape: (178, 13)

k=2  Inertia=1658.76   Silhouette=0.2593
k=3  Inertia=1277.93   Silhouette=0.2849
k=4  Inertia=1175.43   Silhouette=0.2602
k=5  Inertia=1109.51   Silhouette=0.2016
k=6  Inertia=1046.00 Silhouette=0.2372
k=7  Inertia=981.60  Silhouette=0.2036
k=8  Inertia=935.20  Silhouette=0.1570
k=9  Inertia=889.89  Silhouette=0.1499
k=10  Inertia=845.90 Silhouette=0.1436

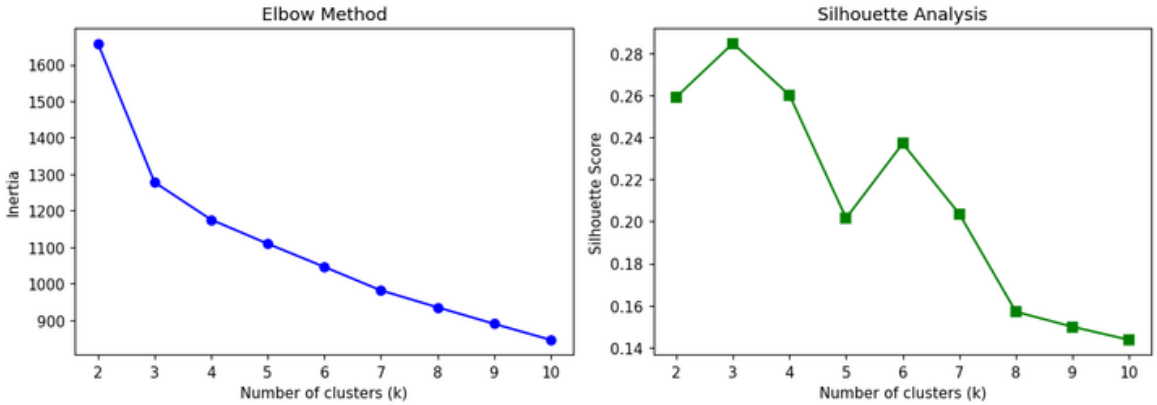
--- K-Means on original 13 features ---
Cluster sizes: [65 51 62]
Silhouette Score: 0.2849
Adjusted Rand Index: 0.8975

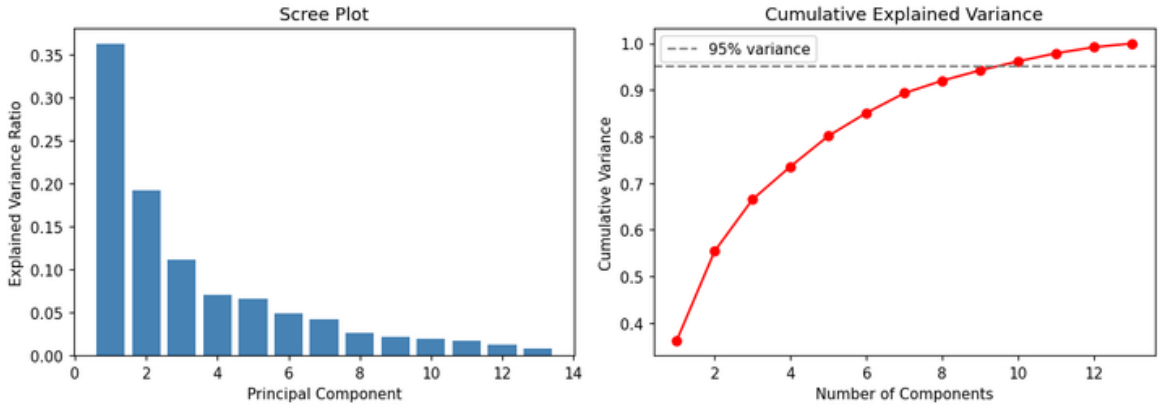
Explained variance ratio (first 5 components): [0.362  0.1921 0.1112 0.0707
    0.0656]
Cumulative variance (first 5 components)     : [0.362  0.5541 0.6653 0.736
    0.8016]
Components needed to retain 95% variance: 10

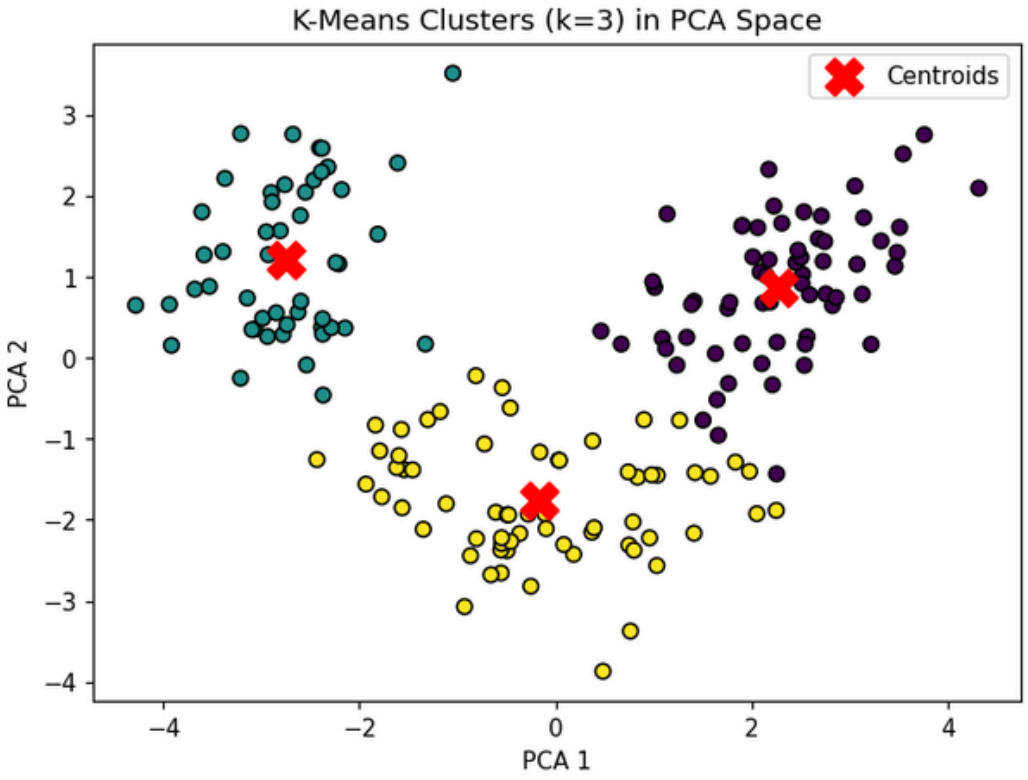
K-Means before and after PCA:

 Data used  Original (13  Silhouette  Adjusted Rand Index
features)     PCA     (10      0.2849               0.8975
components, 95%)   PCA (2      0.2987               0.8975
components)                   0.5611               0.8951

Cluster vs Wine-class cross-tab (PCA 2 components):
  Class    0   1   2
Cluster
0         59   5   0
1          0   1  48
2          0  65   0
```







RESULT:

K-Means clustering and PCA were successfully implemented on the Wine dataset. The best number of clusters was k = 3 (silhouette 0.2849), and the clusters matched the wine classes well (adjusted Rand index 0.8975). PCA showed that 10 of the 13 components retain 95% of the variance, and the first two components retain 55.4%. Clustering the 2-component data gave nearly the same adjusted Rand index (0.8951) with a much higher silhouette score (0.5611), so dimensionality reduction kept the cluster structure while making the data easy to visualise.